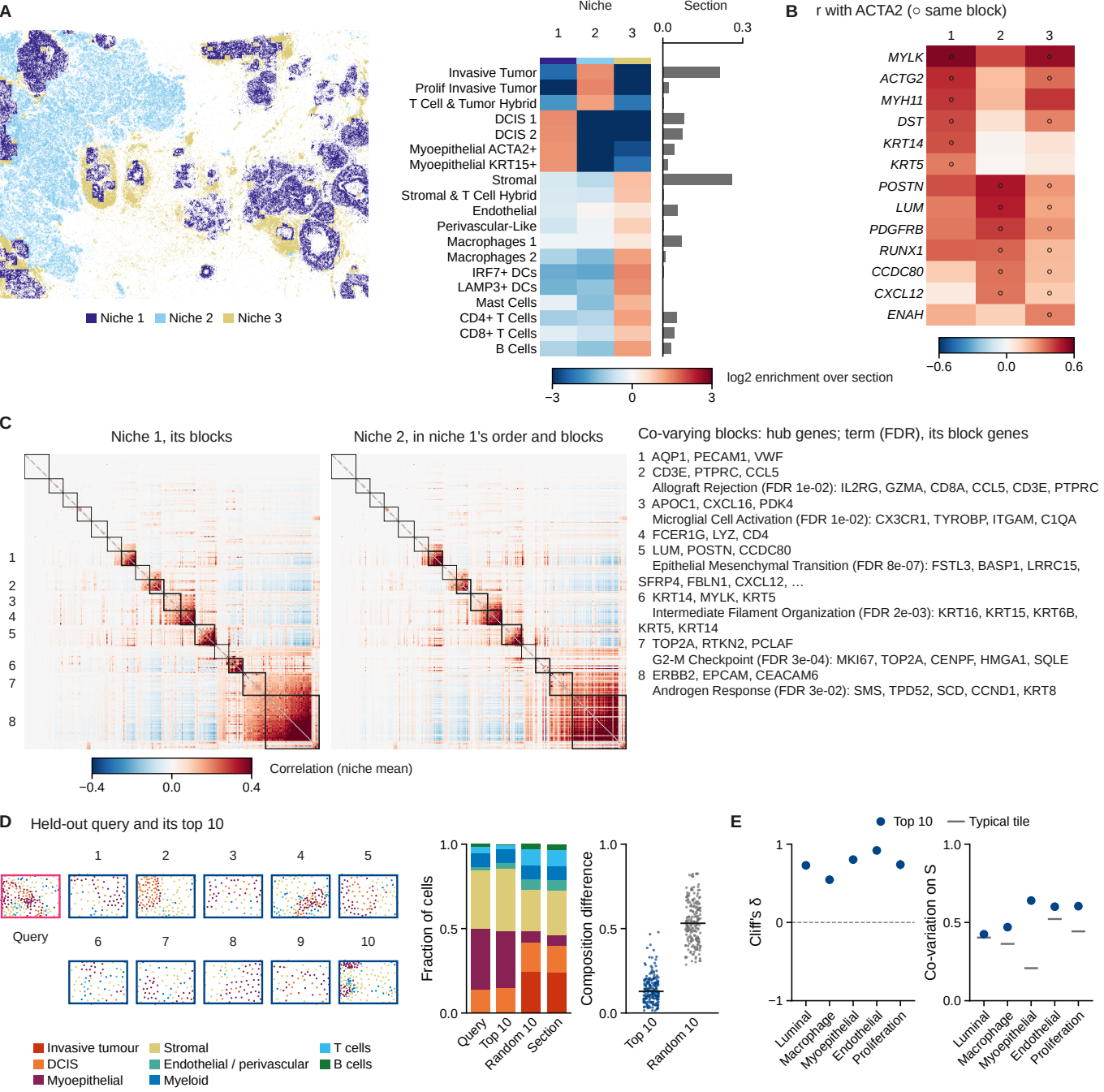

A
Niche 1
Niche 2
Niche 3
Niche
Section
1
2
3
0.0
0.3
Invasive Tumor
Prolif Invasive Tumor
T Cell & Tumor Hybrid
DCIS 1
DCIS 2
Myoepithelial ACTA2+
Myoepithelial KRT15+
Stromal
Stromal & T Cell Hybrid
Endothelial
Perivascular-Like
Macrophages 1
Macrophages 2
IRF7+ DCs
LAMP3+ DCs
Mast Cells
CD4+ T Cells
CD8+ T Cells
B Cells
−3
0
3
log2 enrichment over section
B
r with ACTA2 (○ same block)
1
2
3
MYLK
ACTG2
MYH11
DST
KRT14
KRT5
POSTN
LUM
PDGFRB
RUNX1
CCDC80
CXCL12
ENAH
−0.6
0.0
0.6
C
Niche 1, its blocks
Niche 2, in niche 1's order and blocks
1
2
3
4
5
6
7
8
−0.4
0.0
0.4
Correlation (niche mean)
Co-varying blocks: hub genes; term (FDR), its block genes
1 AQP1, PECAM1, VWF
2 CD3E, PTPRC, CCL5
Allograft Rejection (FDR 1e-02): IL2RG, GZMA, CD8A, CCL5, CD3E, PTPRC
3 APOC1, CXCL16, PDK4
Microglial Cell Activation (FDR 1e-02): CX3CR1, TYROBP, ITGAM, C1QA
4 FCER1G, LYZ, CD4
5 LUM, POSTN, CCDC80
Epithelial Mesenchymal Transition (FDR 8e-07): FSTL3, BASP1, LRRC15, SFRP4, FBLN1, CXCL12, …
6 KRT14, MYLK, KRT5
Intermediate Filament Organization (FDR 2e-03): KRT16, KRT15, KRT6B, KRT5, KRT14
7 TOP2A, RTKN2, PCLAF
G2-M Checkpoint (FDR 3e-04): MKI67, TOP2A, CENPF, HMGA1, SQLE
8 ERBB2, EPCAM, CEACAM6
Androgen Response (FDR 3e-02): SMS, TPD52, SCD, CCND1, KRT8
D
Held-out query and its top 10
1
2
3
4
5
Query
6
7
8
9
10
Invasive tumour
DCIS
Myoepithelial
Stromal
Endothelial / perivascular
Myeloid
T cells
B cells
Fraction of cells
1.0
0.5
0.0
Query
Top 10
Random 10
Section
Composition difference
1.0
0.5
0.0
Top 10
Random 10
E
Top 10
Typical tile
Cliff's δ
1
0
−1
Luminal
Macrophage
Myoepithelial
Endothelial
Proliferation
Co-variation on S
1.0
0.5
0.0
Luminal
Macrophage
Myoepithelial
Endothelial
Proliferation